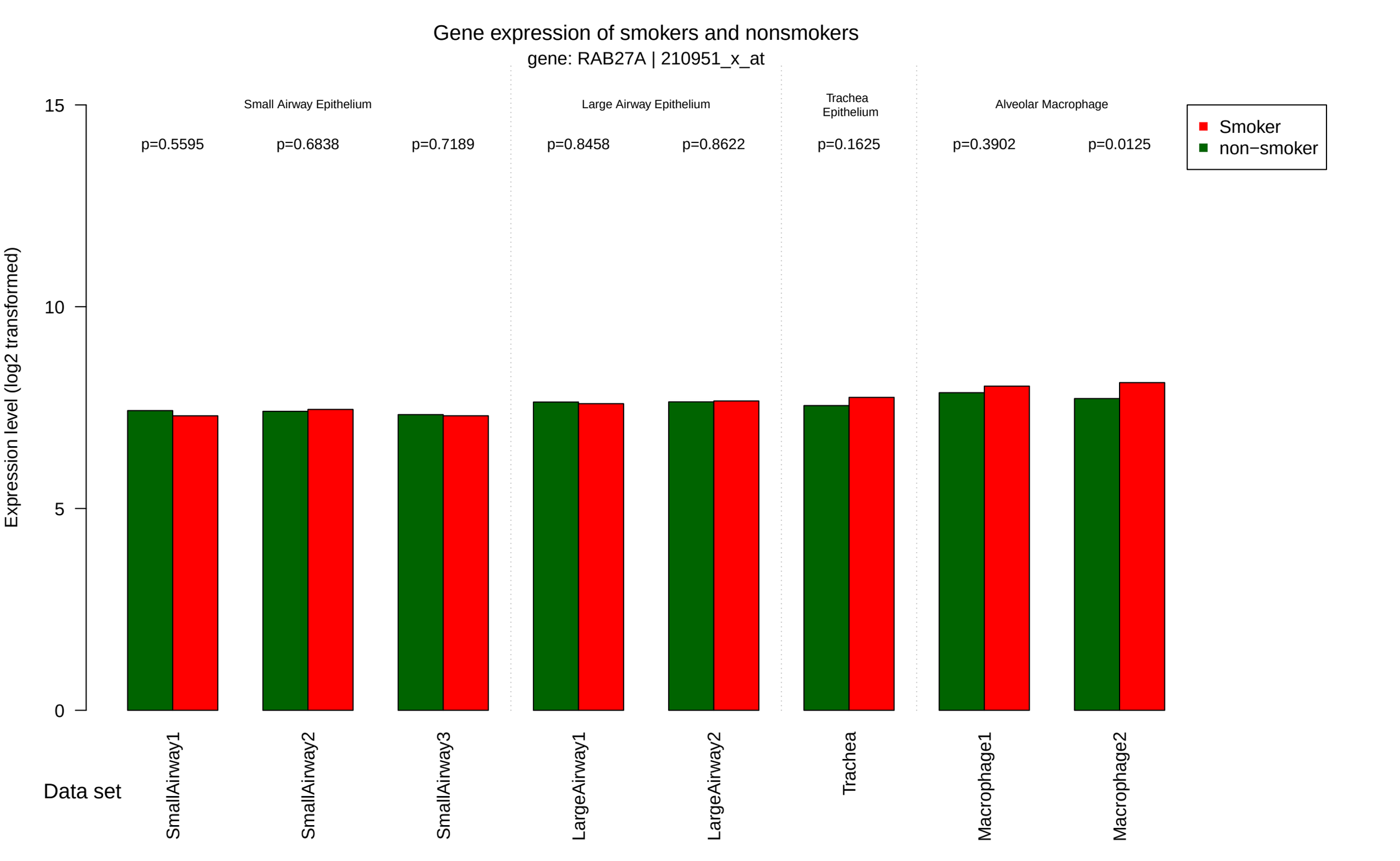
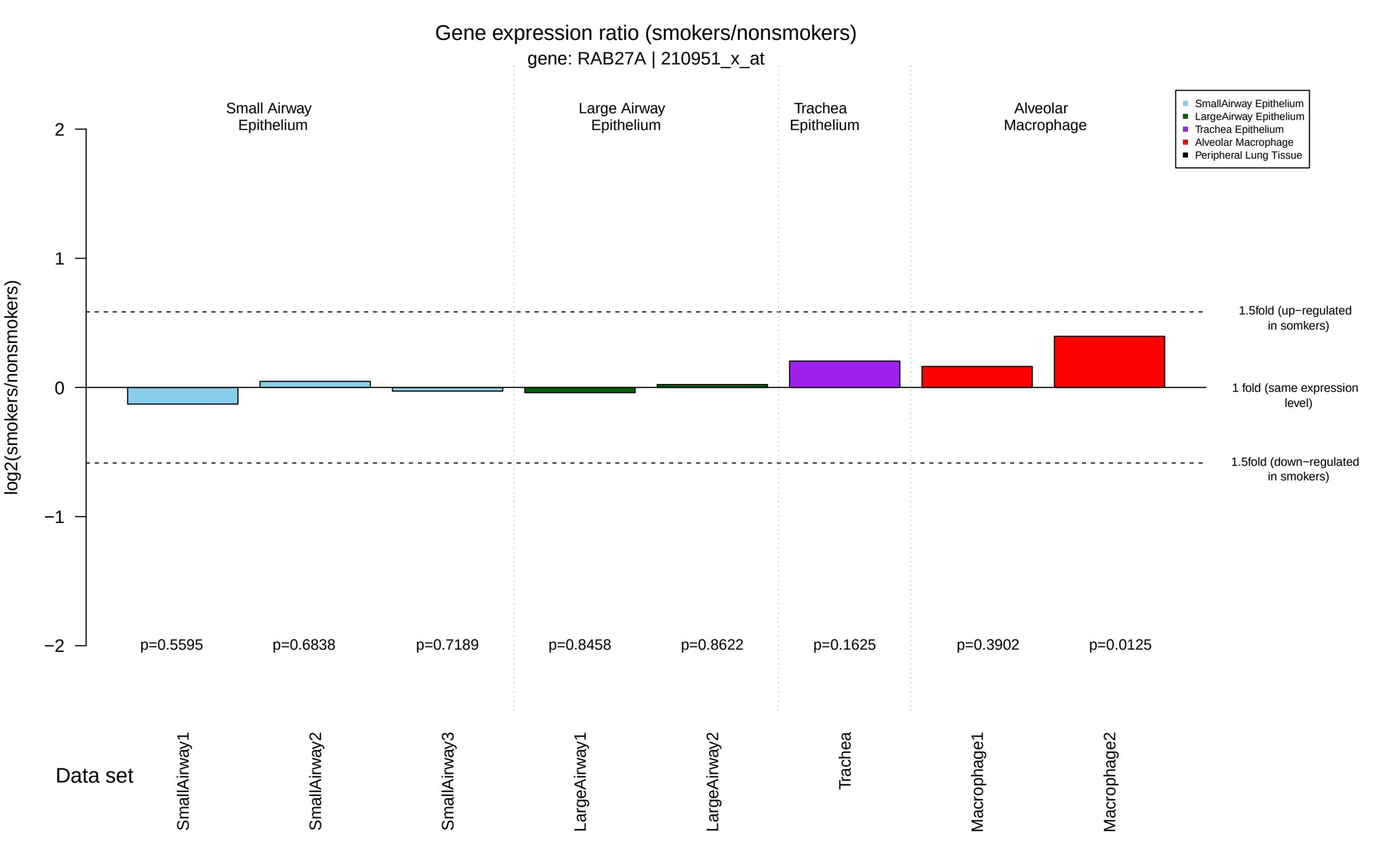
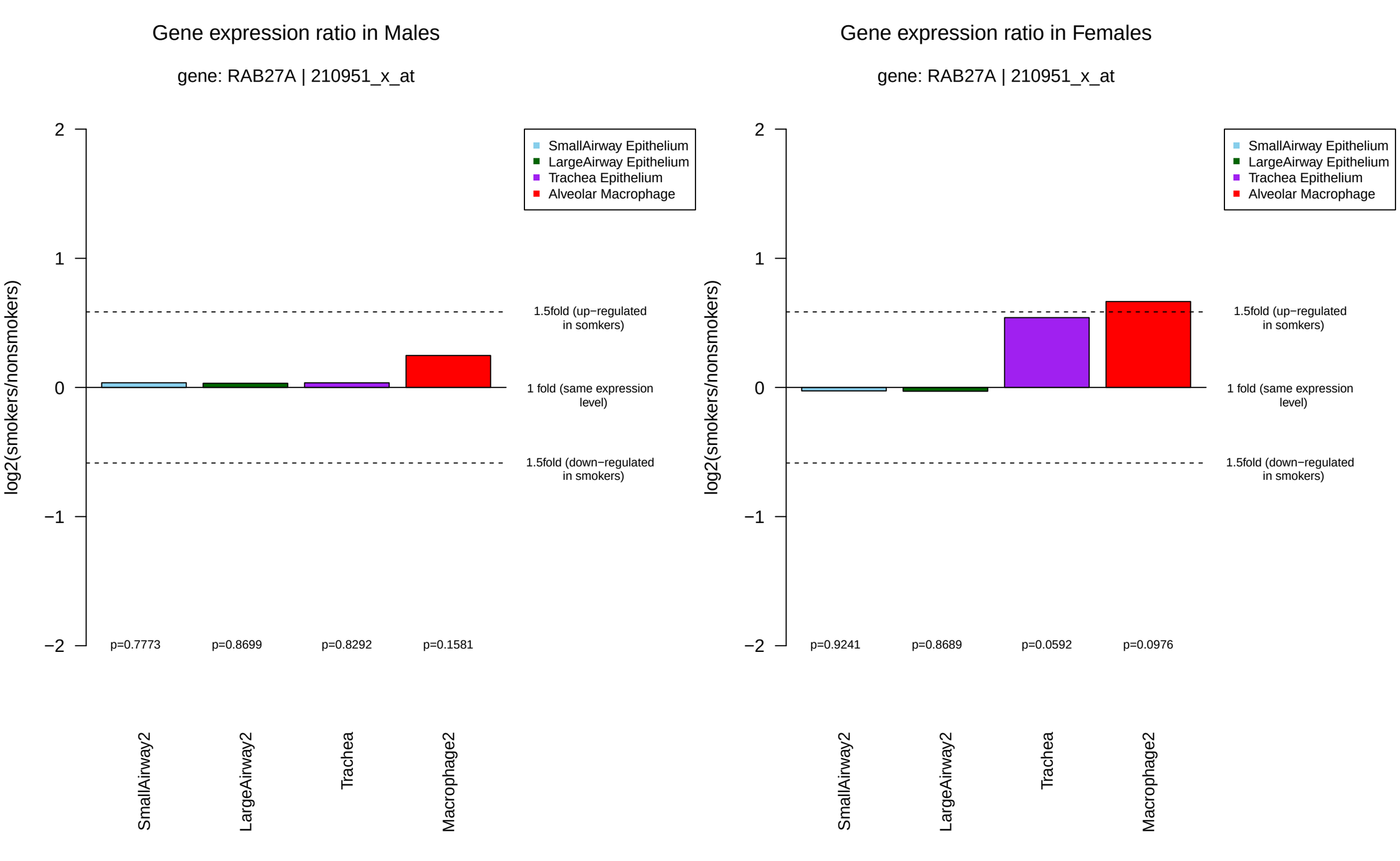
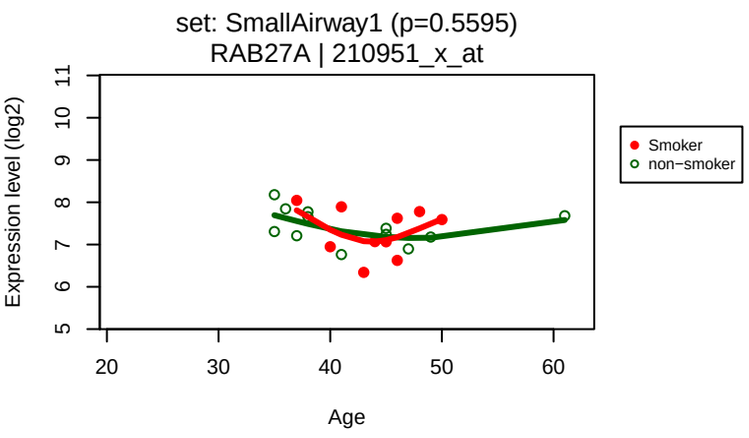
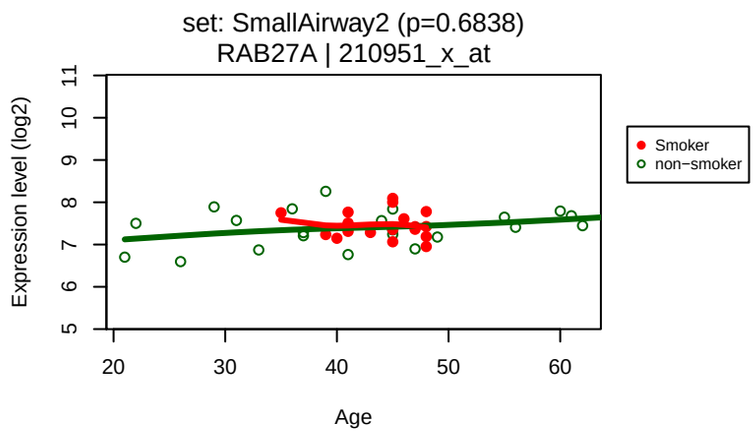
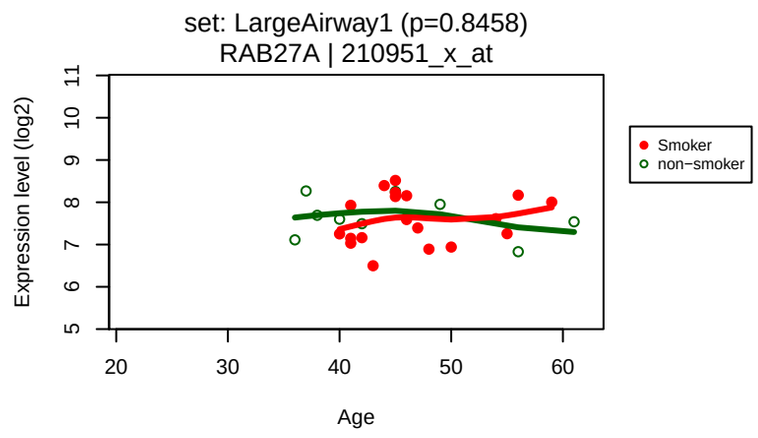

Gene expression of smokers and nonsmokers
gene: RAB27A | 210951_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5595
p=0.6838
p=0.7189
p=0.8458
p=0.8622
p=0.1625
p=0.3902
p=0.0125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

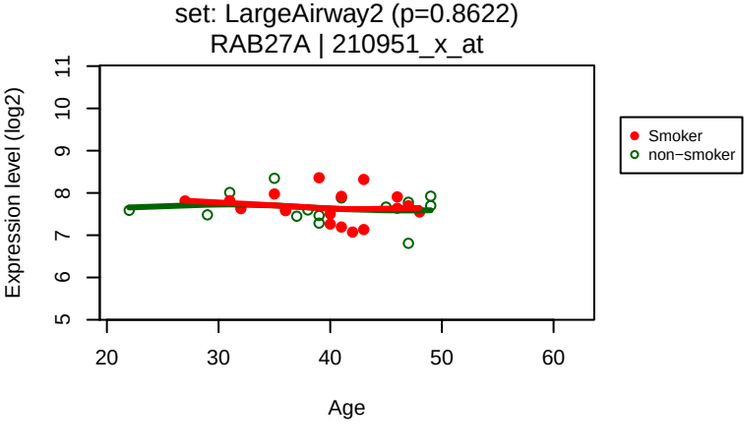

Gene expression ratio (smokers/nonsmokers)
gene: RAB27A | 210951_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5595
p=0.6838
p=0.7189
p=0.8458
p=0.8622
p=0.1625
p=0.3902
p=0.0125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

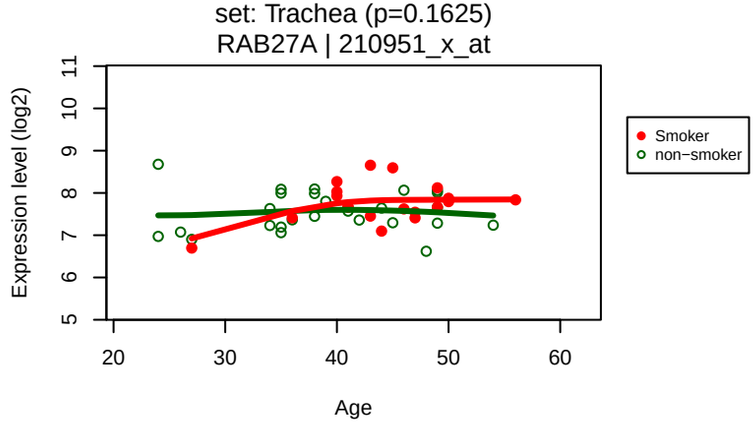

Gene expression ratio in Males
gene: RAB27A | 210951_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7773
p=0.8699
p=0.8292
p=0.1581
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB27A | 210951_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9241
p=0.8689
p=0.0592
p=0.0976
SmallAirway2
LargeAirway2
Trachea
Macrophage2

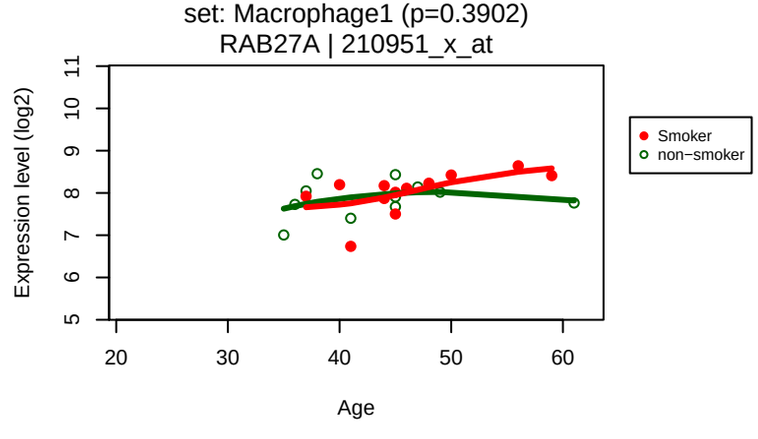

set: SmallAirway1 (p=0.5595)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker

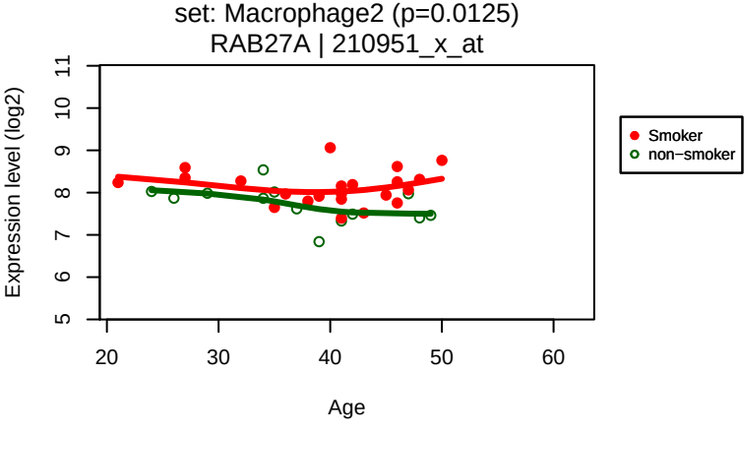

set: SmallAirway2 (p=0.6838)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker

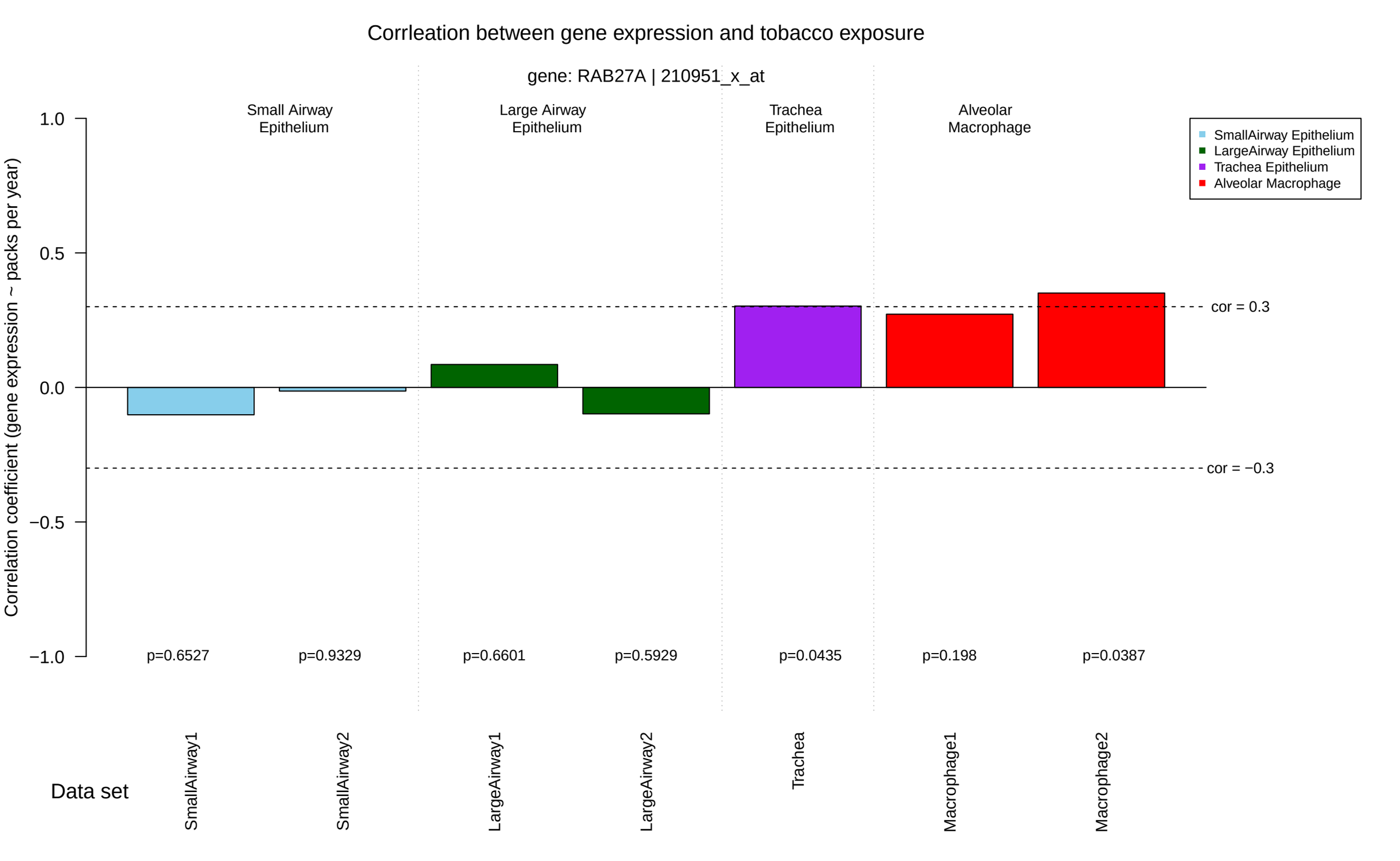

set: LargeAirway1 (p=0.8458)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8622)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1625)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3902)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0125)
RAB27A | 210951_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB27A | 210951_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6527
p=0.9329
p=0.6601
p=0.5929
p=0.0435
p=0.198
p=0.0387
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2